## Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

### 2 Describe one advantage of the MOSFET over the BJT.

- Almost no gate current, so it draws very little current from the controller
- Can switch higher currents with less heat
- Very low resistance when on

### 3 Quick check

- a) To protect the transistor from reverse voltage
- b) Gate voltage
- c) 300 mA
- d) Relay

### 4 Key terms

- 1 – A
- 2 – D
- 3 – E
- 4 – C
- 5 – B
- 6 – F

### 5 Describe what is meant by threshold voltage.

- The gate–source voltage at which the MOSFET starts to conduct (switch on)

**No marks for:** "The voltage to turn it on" with no gate–source

### 6 Explain the purpose of the diode.

- When the coil switches off it produces a large back emf (voltage spike)
- The diode safely conducts this, protecting the transistor from damage

### 7 Explain why a relay is used.

- The transistor circuit can't switch high voltage/current directly
- The relay lets a small current switch a separate high-current circuit safely